

Problem

A certain band-limited periodic voltage has only three harmonics in its Fourier series representation. The harmonics have the following rms values: fundamental 40 V, third harmonic 20 V, fifth harmonic 10 V.

(a) If the voltage is applied across a 5-Ω resistor, find the average power dissipated by the resistor.

(b) If a dc component is added to the periodic voltage and the measured power dissipated increases by 5 percent, determine the value of the dc component added.

Step-by-step solution

Step 1 of 2

$$\begin{aligned} \text{(A)} \quad p &= \frac{V_{DC}^2}{R} + \frac{1}{2} \sum \frac{V_n^2}{R} = \frac{V_{DC}^2}{R} + \sum \frac{V_n^2, rms}{R} \\ &= 0 + \left(\frac{40^2}{5} \right) + \left(\frac{20^2}{5} \right) + \left(\frac{10^2}{5} \right) = 420W \end{aligned}$$

Step 2 of 2

$$\begin{aligned} \text{(B)} \quad 5\% \text{ increase} &= \left(\frac{5}{100} \right) 420 = 21 \\ p_{DC} = 21W &= \frac{V_{DC}^2}{R} \quad \text{which leads to} \\ V_{DC}^2 &= 21R = 105 \\ V_{DC} &= 10.25V \end{aligned}$$